MODULAR KOSZUL DUALITY AND THE SHADOW TOWER: ALGEBRAICITY, DEPTH CLASSIFICATION, AND THE GRAVITATIONAL COPRODUCT

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Abstract. Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X, with modular characteristic $\kappa = \kappa(\mathcal{A})$, cubic shadow α, and quartic shadow S_4. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ defines a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra, whose finite-order projections $\Theta_{\mathcal{A}}^{\leq r}$ form the *shadow Postnikov tower*.

We prove four results. First, on each one-dimensional primary line L in the cyclic deformation complex, the weighted shadow generating function $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ satisfies the algebraic relation $H(t)^2 = t^4 Q(t)$, where $Q(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$ is the *shadow metric*: the tower is algebraic of degree 2 along L, and every coefficient S_r for $r \geq 5$ is determined by the three seed invariants (κ, α, S_4) of L. Second, the *critical discriminant* $\Delta = 8\kappa S_4$ partitions the standard Lie-theoretic landscape into four depth classes: Gaussian ($r_{\max} = 2$), Lie ($r_{\max} = 3$), contact ($r_{\max} = 4$), and mixed ($r_{\max} = \infty$), with $\Delta = 0$ on the primary line forcing finite depth and $\Delta \neq 0$ forcing infinite depth. Third, we compute the genus-2 free energy $F_2 = \kappa \cdot 7/5760$ for single-generator algebras by an explicit stable-graph sum, carry out the first multi-channel computation for $\mathcal{W}_3$ (exhibiting a cross-channel correction whose cancellation is equivalent to multi-generator universality at genus 2), and derive the planted-forest correction $\delta_{\mathrm{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$. Fourth, we prove that the gravitational coproduct, obtained by homological perturbation transfer through BRST reduction from $\widehat{\mathfrak{sl}}_2$ to Virasoro, is strictly primitive at all arities: gravity does not produce particles, only braids. The mechanism (ghost-number obstruction) applies to all principal Drinfeld–Sokolov reductions.

1. Introduction

1.1. **The question.** Classical Koszul duality is a phenomenon of graded algebras over a point. A quadratic algebra A determines a dual coalgebra $A^!$; the bar construction $B(A)$ mediates between them; and the Koszul property (bar cohomology concentrated in bar degree 1) is equivalent to the existence

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of a linear resolution. The theory is the subject of Priddy [15], Beilinson–Ginzburg–Soergel [2], and Loday–Vallette [13].

On an algebraic curve X, the generators of the algebra become sections of a $\mathcal{D}_X$-module, the quadratic relations become operator product expansions with meromorphic singularities, and the bar differential becomes an integral transform whose kernel is the logarithmic 1-form $\eta_{ij} = d\log(z_i - z_j)$ on the Fulton–MacPherson compactification $\overline{C}_n(X)$ of the configuration space. The resulting object, the *chiral bar complex* $\bar{B}^{\mathrm{ch}}(\mathcal{A})$, is a factorization coalgebra on the Ran space $\mathrm{Ran}(X)$ [1, 6, 4].

At genus 0, the classical theory embeds faithfully. At genus $g \geq 1$, the topology of the curve forces genuinely new structure: the bar differential acquires curvature $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the Hodge class on $\overline{\mathcal{M}}_g$ and the scalar $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ is an intrinsic invariant of the algebra called the *modular characteristic*. This curvature has no analogue in the classical setting; its geometric origin is the nontriviality of the Hodge bundle over $\overline{\mathcal{M}}_g$.

The present paper addresses a single question: *what invariants does the chiral bar construction produce, beyond the modular characteristic?*

1.2. **The answer: the shadow tower.** The bar complex on a curve of genus g with n marked points defines a pronilpotent dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ (the modular convolution algebra), whose differential encodes graph sums over stable curves. The boundary-of-boundary relation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ implies $D^2 = 0$ on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$, and the element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is therefore a Maurer–Cartan element: $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$.

The MC element $\Theta_{\mathcal{A}}$ encodes the full genus expansion. Its finite-order projections $\Theta_{\mathcal{A}}^{\leq r}$, obtained by truncating the arity filtration at level r, form the *shadow Postnikov tower*. At arity 2, the tower produces the modular characteristic κ. At arity 3, a cubic shadow α (the structure constant of the transferred L_∞ bracket ℓ_3). At arity 4, a quartic shadow S_4 (the first genuinely nonlinear invariant, measuring the contact term in the four-point collision residue).

The obstruction to truncating $\Theta_{\mathcal{A}}$ at finite arity is measured by classes $o_{r+1}(\mathcal{A}) \in H^2(\mathcal{A}_{r+1,0}^{\mathrm{sh}})$ in the shadow algebra $\mathcal{A}^{\mathrm{sh}} := H_\bullet(\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A}))$. When $o_{r+1} = 0$, the tower stops; when $o_{r+1} \neq 0$, the next shadow coefficient S_{r+1} is forced. The question is how far this process continues and what controls it.

1.3. **The shadow metric and algebraicity.** The first main result (Theorem 4.1, parts (i)–(ii)) is that the shadow tower is controlled by a single quadratic polynomial.

Restrict to a one-dimensional primary slice L of the cyclic deformation complex (the space spanned by a single primary field η of definite conformal weight). On L, the MC recursion reduces to a scalar recursion for the shadow coefficients S_r, and the weighted generating function $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$

satisfies the algebraic relation

$$H(t)^2 \;=\; t^4\, Q(t), \tag{1.1}$$

where the *shadow metric*

$$Q(t) \;=\; 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2 \tag{1.2}$$

is a quadratic polynomial whose coefficients are determined by the three seed invariants (κ, α, S_4) of the algebra. The generating function is therefore algebraic of degree 2: $H(t) = t^2\sqrt{Q(t)}$. Every shadow coefficient S_r for $r \geq 5$ is a rational function of (κ, α, S_4), computable from the binomial expansion of $\sqrt{Q}$.

The proof converts the Maurer–Cartan recursion into a convolution identity for the Taylor coefficients of $F(t) = H(t)/t^2$, showing $F^2 = Q$. The orders $m = 0, 1, 2$ of this identity reproduce the definitions of κ, α, S_4; the orders $m \geq 3$ are exactly the recursion.

This is the *intrinsic quartic principle*: an infinite sequence of invariants, one for each arity $r \geq 2$, is determined by three numbers. The situation is analogous to the role of the Weierstrass equation for an elliptic curve: two parameters (g_2, g_3) determine all Laurent coefficients of $\wp$, because $\wp$ satisfies an algebraic equation of degree 2 in $\wp'$. Here, three parameters (κ, α, S_4) determine all shadow coefficients S_r, because H satisfies an algebraic equation of degree 2 in H.

1.4. **The four-class classification.** Write $Q(t) = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2$ for the *Gaussian decomposition*, where $\Delta := 8\kappa S_4$ is the *critical discriminant*. The Gaussian envelope $(2\kappa + 3\alpha t)^2$ is the part of Q visible to the quadratic and cubic shadows alone; Δ measures the irreducible quartic contribution.

The critical discriminant controls a fundamental dichotomy.

When $\Delta = 0$, the shadow metric is a perfect square: $Q = (2\kappa + 3\alpha t)^2$, the generating function $H(t) = t^2(2\kappa + 3\alpha t)$ is polynomial, and the tower terminates. If $\alpha = 0$, only $S_2 = \kappa$ survives (depth 2); if $\alpha \neq 0$, then S_2 and S_3 survive (depth 3).

When $\Delta \neq 0$, the shadow metric is irreducible: $\sqrt{Q}$ is irrational, every Taylor coefficient of $\sqrt{Q}$ beyond degree 1 carries a factor of Δ, and $S_r \neq 0$ for all $r \geq 4$ (generically). The tower is infinite.

This yields the second main result (Theorem 4.1, part (iii)): every chirally Koszul algebra whose primary-line shadow data has been computed falls into one of four *shadow depth classes*. The partition is exhaustive for the standard Lie-theoretic landscape (Table **??**); whether every chirally Koszul algebra belongs to one of these four classes is a structural observation, not a classification theorem.

Class	Δ	$r_{\max}$	Prototype	Shadow data
G (Gaussian)	0	2	Heisenberg	$\kappa = k$
L (Lie)	0	3	Affine Kac–Moody	κ, $\alpha \neq 0$
C (Contact)	0^*	4	$\beta\gamma$ system	$\kappa = 6\lambda^2 - 6\lambda + 1$
M (Mixed)	$\neq 0$	∞	Virasoro, $\mathcal{W}_N$	κ, α, S_4

*Class **C**: $\Delta = 0$ on the primary line; the quartic contact invariant lives on a charged stratum.

Class **G** consists of algebras whose bar complex is formal in the classical sense: the shadow tower terminates at the leading scalar, and all higher-genus free energies F_g are determined by κ alone. Class **L** adds exactly one cubic correction (the structure constant of the Lie bracket on generators), after which the tower terminates by the Jacobi identity. Class **C** is the first nonlinear class: the quartic contact invariant S_4 is nonzero but lives on a separate stratum whose self-bracket exits the complex, so the tower terminates after one further step. Class **M** is the generic class: the shadow tower is infinite, the free energies F_g receive corrections at every arity, and the shadow coefficients S_r decay (or grow) at a computable rate governed by the *shadow growth rate*

$$\rho(\mathcal{A}) \;=\; \sqrt{\frac{9\alpha^2 + 16\kappa S_4}{4\kappa^2}}. \tag{1.3}$$

The growth rate is the reciprocal of the branch-point modulus of the shadow metric. When $\rho < 1$, the shadow tower converges absolutely; when $\rho > 1$, it diverges but the algebraic function $\sqrt{Q}$ extends analytically past the convergence radius. For the Virasoro algebra at central charge c:

$$\rho(\mathrm{Vir}_c) \;=\; \frac{1}{c}\sqrt{\frac{180c + 872}{5c + 22}}, \tag{1.4}$$

with a critical central charge $c_\star \approx 6.125$ (the unique positive root of $5c^3 + 22c^2 - 180c - 872 = 0$) separating the convergent and divergent regimes. At the self-dual point $c = 13$ (where $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ satisfies $\mathrm{Vir}_c = \mathrm{Vir}_c^!$): $\rho \approx 0.467$.

Classical Koszul duality over a point is the genus-0, arity-2, $\Delta = 0$ stratum of this picture: the formal Gaussian case where the tower terminates immediately. Everything beyond class **G** is new.

1.5. **The shadow connection and Koszul monodromy.** The shadow metric equips each primary line with a logarithmic connection

$$\nabla^{\mathrm{sh}} \;=\; d \;-\; \frac{Q'(t)}{2\,Q(t)}\,dt, \tag{1.5}$$

with regular singularities at the zeros of Q and residue $\frac{1}{2}$. The monodromy around each branch point is -1. This is the *Koszul sign*: the square root

$\sqrt{Q}$ changes sign around a simple zero, and the shadow tower inherits this sign change through the algebraic relation $H = t^2\sqrt{Q}$.

Under Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$, the shadow metric transforms by the Verdier involution. For the Virasoro family $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$, the discriminants satisfy

$$\Delta(\mathrm{Vir}_c) + \Delta(\mathrm{Vir}_{26-c}) \;=\; \frac{6960}{(5c+22)(152-5c)}, \tag{1.6}$$

a Virasoro-specific identity whose numerator $6960 = 40{\cdot}174$ is a product of the Virasoro quartic normalization and the sum of the Lee–Yang denominators. The Koszul dual singularities $c = -22/5$ and $c = 152/5$ sum to 26: they are Koszul partners. For other families (affine Kac–Moody, $\mathcal{W}_N$ with $N \geq 3$), the discriminant sum has a different numerator.

1.6. **Genus-2 free energies.** The shadow tower produces invariants of moduli spaces by evaluating graph sums over stable curves. The genus-g *free energy* $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ is computed for each chirally Koszul algebra by the *Faber–Pandharipande integral* $\lambda_g^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_g} \lambda_g$, where $\lambda_g = c_g(\mathbb{E})$ is the top Chern class of the Hodge bundle.

At genus 2: $\lambda_2^{\mathrm{FP}} = 7/5760$, and $F_2(\mathcal{A}) = \kappa \cdot 7/5760$ for every single-generator chirally Koszul algebra. The verification requires an explicit sum over the six stable graphs of genus 2 with no marked points (the smooth genus-2 curve, figure-eight, banana, dumbbell, theta, and lollipop), weighted by automorphism factors and vertex contributions from the shadow tower.

For the $\mathcal{W}_3$ algebra, which has two generators T (weight 2) and W (weight 3) producing two primary lines, the genus-2 computation is a multi-channel graph sum: each edge of the stable graph carries a channel label (T or W), each vertex contributes a shadow coefficient on the corresponding primary line, and the propagator $P_T = 2/c$, $P_W = 3/c$ depends on the channel. The per-channel diagonal sum yields $F_2^{\mathrm{scal}}(\mathcal{W}_3) = \kappa \cdot \lambda_2^{\mathrm{FP}} = 7c/6912$, but the full multi-channel graph sum produces an additional cross-channel correction $\delta F_2 = (c+120)/(16c)$. Whether this correction is cancelled by CohFT R-matrix terms (restoring the universality identity $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ at genus 2 for multi-generator algebras) is the content of the multi-generator universality problem, which remains open.

The Maurer–Cartan equation on $\overline{\mathcal{M}}_{2,0}$ also produces a *planted-forest correction*

$$\delta_{\mathrm{pf}}^{(2,0)} \;=\; \frac{S_3(10\,S_3 - \kappa)}{48}, \tag{1.7}$$

measuring the contribution of codimension-2 boundary strata to the genus-2 amplitude. This correction vanishes for class **G** (where $S_3 = 0$). For classes **C** and **M**, it is generically nonzero. For the Virasoro algebra: $\delta_{\mathrm{pf}}^{(2,0)} = -(c-40)/48$.

1.7. **The gravitational coproduct.** The fourth result concerns the transferred algebraic structure under BRST reduction. The Drinfeld–Sokolov

(DS) reduction functor takes affine Kac–Moody data to $\mathcal{W}$-algebra data via a BRST complex with ghosts. The homological perturbation lemma (HPL) transfers the A_∞ products and the coproduct from the affine dg-shifted Yangian to a transferred Yangian on the $\mathcal{W}$-algebra sector.

We prove the result for the principal DS reduction $\widehat{\mathfrak{sl}}_2 \to \mathrm{Vir}$. The ghost-number obstruction (the homotopy h in the SDR has ghost degree -1, and the projection $p \otimes p$ kills all ghost-number-(-1) elements) forces the transferred coproduct to be *strictly primitive at all arities*. The mechanism applies to any principal DS reduction $\widehat{\mathfrak{g}} \to \mathcal{W}(\mathfrak{g})$, since the ghost-number grading of the BRST complex is universal; we state the general case as Conjecture 9.2 and prove it for $\widehat{\mathfrak{sl}}_2$:

$$\Delta_z^{\mathrm{Vir}}(T) \;=\; \tau_z(T) \otimes 1 + 1 \otimes T, \qquad \Delta_{z,n}^{\mathrm{Vir}} = 0 \quad \text{for } n \geq 2.$$

All higher-arity coproduct corrections, which would represent graviton splitting into constituent gravitons, are killed by the ghost-number mechanism. The only nontrivial gravitational structure is the r-matrix twist $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ on the target product, encoding the braiding of line operators.

The physical content: in Chern–Simons gauge theory, gluons split (the coproduct is nontrivial, reflecting the Clebsch–Gordan decomposition of tensor products). In 3d gravity, gravitons do not split. The gravitational coproduct is primitive; all interactions are encoded in the twist, not in particle production.

1.8. **Relation to existing work.** The bar construction for chiral algebras was introduced by Beilinson and Drinfeld [1] and developed in the factorization algebra framework by Francis and Gaitsgory [6] and Costello and Gwilliam [4]. The modular characteristic κ appears implicitly in the work of Feigin and Fuchs on Hodge bundles of conformal blocks, and the formula $\kappa(\mathrm{Vir}_c) = c/2$ can be extracted from Beilinson and Schechtman [3]. The invariants of moduli spaces produced by vertex algebras via conformal blocks are the subject of Faber and Pandharipande [7], Mumford [14], and the extensive literature on cohomological field theories following Kontsevich and Manin [10].

The quartic shadow S_4, the critical discriminant Δ, the four-class partition, and the algebraicity theorem (Theorem 4.1) appear to be new. The connection between shadow depth and L_∞ non-formality, while consonant with the general philosophy of Kontsevich [11], has not been established previously in the chiral setting.

The Drinfeld–Sokolov reduction and its homological algebra are classical [12, 8, 5]; the gravitational coproduct primitivity result (Theorem 9.1) and its connection to the shadow tower appear to be new.

1.9. **Structure of the paper.** Section 2 reviews chiral algebras, the bar construction, and the modular characteristic, establishing notation and the three seed invariants (κ, α, S_4). Section 3 defines the modular convolution

algebra and the Maurer–Cartan element $\Theta_{\mathcal{A}}$. Section 4 proves the Riccati algebraicity theorem and derives the shadow metric, the closed-form generating function, and the shadow connection. Section 5 proves the four-class classification and the universal factorization theorem. Section 6 computes (κ, α, S_4) for every standard family and presents explicit shadow towers. Section 7 carries out the genus-2 graph sum, including the multi-channel computation for $\mathcal{W}_3$ and the planted-forest correction. Section 8 develops the complementarity constraints and duality identities. Section 9 proves the gravitational coproduct primitivity. Section 10 discusses open problems, including multi-generator universality at genus ≥ 3, the operadic complexity conjecture, the connection to Pixton's ideal, and the arithmetic content of shadow denominators.

2. Chiral Koszul pairs and the bar construction

(To be written.)

3. The modular convolution algebra

(To be written.)

4. The shadow metric and Riccati algebraicity

Theorem 4.1 (Main theorem: algebraicity, classification, and genus-2 formula)**.** *Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X. Let $\kappa = \kappa(\mathcal{A})$ be the modular characteristic, and let α, S_4 be the cubic and quartic shadow invariants. Define the* shadow metric

$$Q(t) \;=\; 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$$

and the critical discriminant $\Delta = 8\kappa S_4$.

(i) ***Algebraicity.*** *The weighted shadow generating function $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ satisfies*

$$H(t)^2 = t^4\, Q(t).$$

Every shadow coefficient S_r for $r \geq 5$ is determined by (κ, α, S_4) via $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q(t)}$.

(ii) ***Shadow connection.*** *The shadow metric defines a logarithmic connection $\nabla^{\mathrm{sh}} = d - Q'/(2Q)\, dt$ with residue $\frac{1}{2}$ at each zero of Q. Monodromy $= -1$ (Koszul sign).*

(iii) ***Four-class partition.*** *Write $Q = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2$. The shadow depth $r_{\max}$ lies in $\{2, 3, 4, \infty\}$:*

- *Class* **G***: $\Delta = 0$, $\alpha = 0$, $r_{\max} = 2$.*
- *Class* **L***: $\Delta = 0$, $\alpha \neq 0$, $r_{\max} = 3$.*
- *Class* **C***: $\Delta = 0$ on the primary line; quartic contact on a charged stratum, $r_{\max} = 4$.*
- *Class* **M***: $\Delta \neq 0$, $r_{\max} = \infty$.*

(iv) ***Universal factorization.*** *For $r \geq 4$: $S_r = \Delta \cdot R_r$, where $R_r \in \mathbb{Q}(\alpha, \kappa, S_4)$ is independent of the algebra.*

(v) ***Planted-forest formula.*** *The genus-2 planted-forest correction on a primary line is $\delta_{\mathrm{pf}}^{(2,0)} = S_3(10\,S_3 - \kappa)/48$. It vanishes for class* **G** *(where $S_3 = 0$).*

Proof. (To be written.) □

5. The four-class classification

(To be written.)

6. Shadow towers for the standard landscape

(To be written.)

7. Genus-2 free energies

(To be written.)

8. Complementarity and duality constraints

(To be written.)

9. The gravitational coproduct

Theorem 9.1 (Gravitational primitivity)**.** *Let Δ_z^{aff} be the coproduct of the dg-shifted Yangian $Y^{\mathrm{dg}}(\widehat{\mathfrak{sl}}_2)$, and let (i, p, h) be the SDR from the BRST complex to the Virasoro cohomology. The HPL-transferred coproduct satisfies*

$$\Delta_z^{\mathrm{Vir}}(T) \;=\; \tau_z(T) \otimes 1 + 1 \otimes T.$$

All higher-arity coproduct corrections vanish: $\Delta_{z,n}^{\mathrm{Vir}} = 0$ for $n \geq 2$.

Proof. (To be written.) □

Conjecture 9.2 (General DS primitivity)**.** *For any simple Lie algebra $\mathfrak{g}$, the HPL-transferred coproduct of the principal $\mathcal{W}$-algebra $\mathcal{W}(\mathfrak{g})$ is strictly primitive at all arities. The ghost-number obstruction of the BRST complex for the principal Drinfeld–Sokolov reduction $V_k(\mathfrak{g}) \to \mathcal{W}_k(\mathfrak{g})$ forces $\Delta_{z,n}^{\mathcal{W}} = 0$ for $n \geq 2$.*

10. Open problems and outlook

(To be written.)

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